

SYNTRIX — Looker Studio Dashboard Design & Implementation Guide

Project: CloudMart Enterprise Log Monitoring & Visualization

Version: 1.0

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1. Executive Summary

1.1 Project Objective

Build a **cloud-based log monitoring and visualization solution** that gathers and analyzes application logs from the CloudMart microservices platform to help development teams:

- Track system performance (latency, throughput, uptime)
- Identify error patterns (4xx client errors, 5xx server errors)
- Monitor overall cloud infrastructure health across Cloud Run, GKE, and Compute Engine
- Suggest cost-saving actions based on traffic analysis

1.2 What This Document Delivers

A **complete, beginner-friendly, step-by-step guide** to building a professional, multi-layer Looker Studio dashboard with:

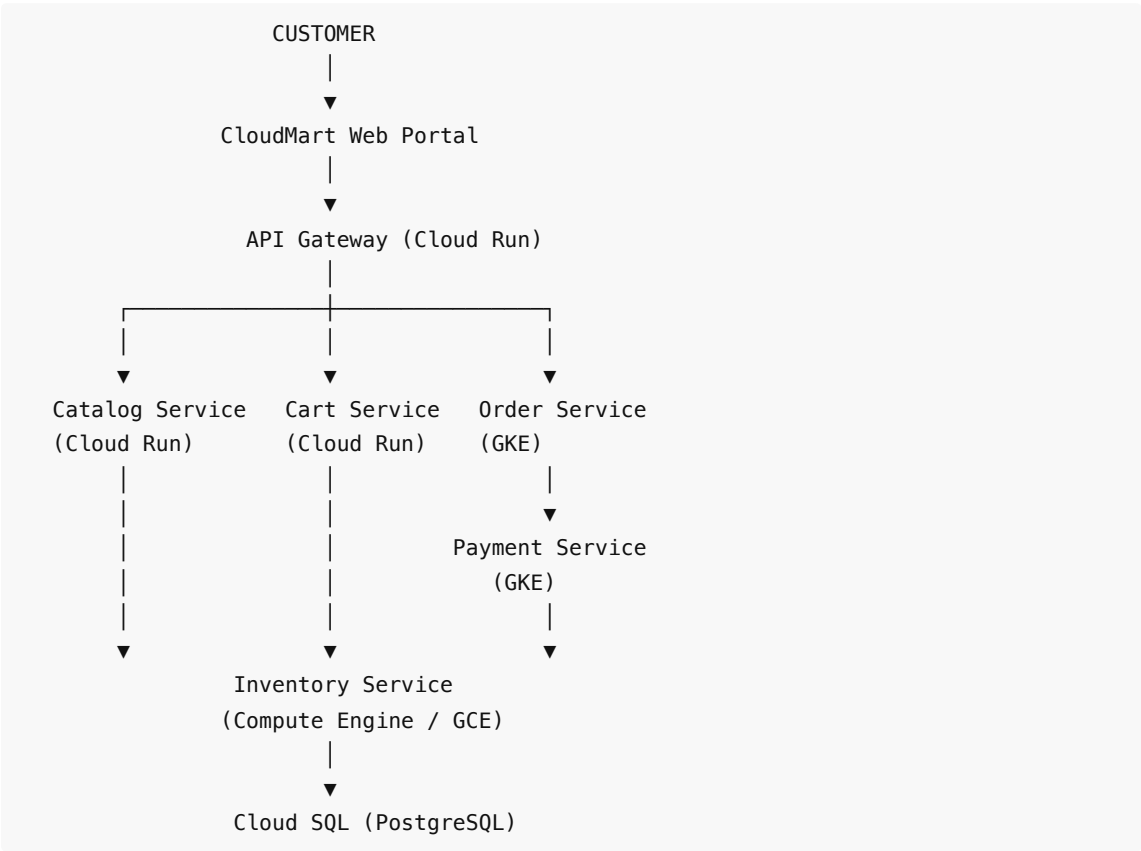
- 6 dedicated dashboard sheets (tabs) for different stakeholder perspectives
- A left-side navigation panel for seamless sheet-to-sheet navigation
- Responsive design optimized for 1920×1080 and laptop screens
- Real GCP BigQuery data connections

1.3 Target Audience

Stakeholder	Dashboard Sheet
CTO / VP Engineering	Sheet 1 — Executive Overview
SRE / DevOps	Sheet 2 — Performance, Sheet 3 — Errors
FinOps / Cloud Mgmt	Sheet 4 — Infrastructure, Sheet 5 — Cost
Developers / QA	Sheet 6 — Log Explorer

2. Architecture & Data Pipeline Overview

2.1 Application Architecture



2.2 Compute Environment Mapping

Service	GCP Compute Platform	Resource Type in Logs
API Gateway	Cloud Run	cloud_run_revision
Catalog Service	Cloud Run	cloud_run_revision
Cart Service	Cloud Run	cloud_run_revision
Order Service	GKE	k8s_container
Payment Service	GKE	k8s_container

Inventory Service	Compute Engine	gce_instance
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2.3 Data Pipeline



2.4 Log Schema (JSON Payload Fields)

Each log entry from services contains these key fields:

Field	Type	Description
timestamp	STRING	ISO 8601 timestamp of the log entry
severity	STRING	Log level: INFO, WARNING, ERROR
event	STRING	Event type: request_completed, order_created, payment_processed, etc.
endpoint	STRING	API endpoint path (e.g., /api/products)
method	STRING	HTTP method: GET, POST, PUT, DELETE
status_code	INT	HTTP response status code
latency_ms	INT	Request processing time in milliseconds
error	STRING	Error message (if applicable)
request_id	STRING	Distributed tracing: unique per request
trace_id	STRING	Distributed tracing: spans entire transaction
service	STRING	Originating service name
order_id	INT	Order identifier (for order/payment events)
total_amount	FLOAT	Transaction amount (for order events)

3. Key Performance Indicators (KPIs)

3.1 Primary KPIs (Executive Level)

KPI	Formula / Source	Target
Global Success Rate	<code>(Requests with status < 400) / Total Requests × 100</code>	≥ 99.5%
Global Error Rate	<code>(Requests with status ≥ 400) / Total Requests × 100</code>	≤ 0.5%
Total Request Volume (24h)	<code>COUNT(*)</code> from cleaned logs, last 24 hours	Trend ↑
System Availability	Uptime % based on error-free intervals	≥ 99.9%

3.2 Performance KPIs (DevOps Level)

KPI	Formula / Source	Threshold
Average Latency	<code>AVG(latency_ms)</code> per service	≤ 200ms
P50 Latency (Median)	<code>APPROX_QUANTILES(latency_ms, 100)[OFFSET(50)]</code>	≤ 100ms
P90 Latency	<code>APPROX_QUANTILES(latency_ms, 100)[OFFSET(90)]</code>	≤ 300ms
P99 Latency (Tail)	<code>APPROX_QUANTILES(latency_ms, 100)[OFFSET(99)]</code>	≤ 1000ms
Max Latency	<code>MAX(latency_ms)</code> per service	≤ 5000ms

3.3 Error KPIs (Reliability Level)

KPI	Formula / Source	Threshold
Client Error Rate (4xx)	<code>COUNTIF(status_code BETWEEN 400 AND 499) / Total</code>	≤ 2%
Server Error Rate (5xx)	<code>COUNTIF(status_code >= 500) / Total</code>	≤ 0.5%
Error Rate by Service	Per-service error breakdown	Per SLA
Error Rate by Environment	Cloud Run vs GKE vs GCE	Per SLA

3.4 Infrastructure & Cost KPIs

KPI	Formula / Source	Action
Requests per Environment	<code>COUNT(*)</code> grouped by <code>compute_environment</code>	Monitor utilization
Daily Request Trend	<code>COUNT(*)</code> grouped by <code>DATE(timestamp)</code>	Capacity planning
Cost Optimization Flags	Rules-based analysis (low-traffic GKE → Cloud Run)	Actionable savings

4. BigQuery Data Sources & Views

These 6 pre-built BigQuery views power the Looker Studio dashboards. They live in the `syntrix_logs` dataset.

View 1: `looker_cleaned_logs` (Base View)

Purpose: Normalized, cleaned base table for all downstream analysis.

Column	Type	Extracted From
timestamp	TIMESTAMP	Raw log timestamp
severity	STRING	Raw log severity
compute_environment	STRING	resource.type
service_name	STRING	Derived from resource labels
endpoint	STRING	jsonPayload.endpoint
method	STRING	jsonPayload.method
status_code	INT64	jsonPayload.status_code
latency_ms	INT64	jsonPayload.latency_ms
event_type	STRING	jsonPayload.event
error_message	STRING	jsonPayload.error

View 2: `looker_latency_analysis`

Purpose: Hourly latency aggregation by service and endpoint.

Column	Type	Description
time_window	TIMESTAMP	Truncated to hour
service_name	STRING	Service identifier
compute_environment	STRING	GCP platform type
endpoint	STRING	API path
total_requests	INT64	Count of requests
avg_latency_ms	FLOAT64	Average response time
max_latency_ms	INT64	Maximum response time

View 3: `looker_error_analysis`

Purpose: Hourly error breakdown with 4xx/5xx split.

Column	Type	Description
time_window	TIMESTAMP	Truncated to hour

service_name	STRING	Service identifier
compute_environment	STRING	GCP platform type
total_requests	INT64	Count of requests
client_errors_4xx	INT64	Client error count
server_errors_5xx	INT64	Server error count
error_rate_percentage	FLOAT64	Overall error rate %

View 4: `looker_usage_monitoring`

Purpose: Daily request volume per compute environment and endpoint.

Column	Type	Description
reporting_date	TIMESTAMP	Truncated to day
compute_environment	STRING	GCP platform type
endpoint	STRING	API path
daily_requests	INT64	Count of daily requests

View 5: `looker_performance_kpis`

Purpose: Advanced latency percentiles (P50/P90/P99) and success rates by service.

Column	Type	Description
service_name	STRING	Service identifier
compute_environment	STRING	GCP platform type
total_requests	INT64	Total request count
p50_latency	INT64	50th percentile latency
p90_latency	INT64	90th percentile latency
p99_latency	INT64	99th percentile latency
success_rate_percentage	FLOAT64	% of requests < 400

View 6: `looker_cost_optimization_recommendations`

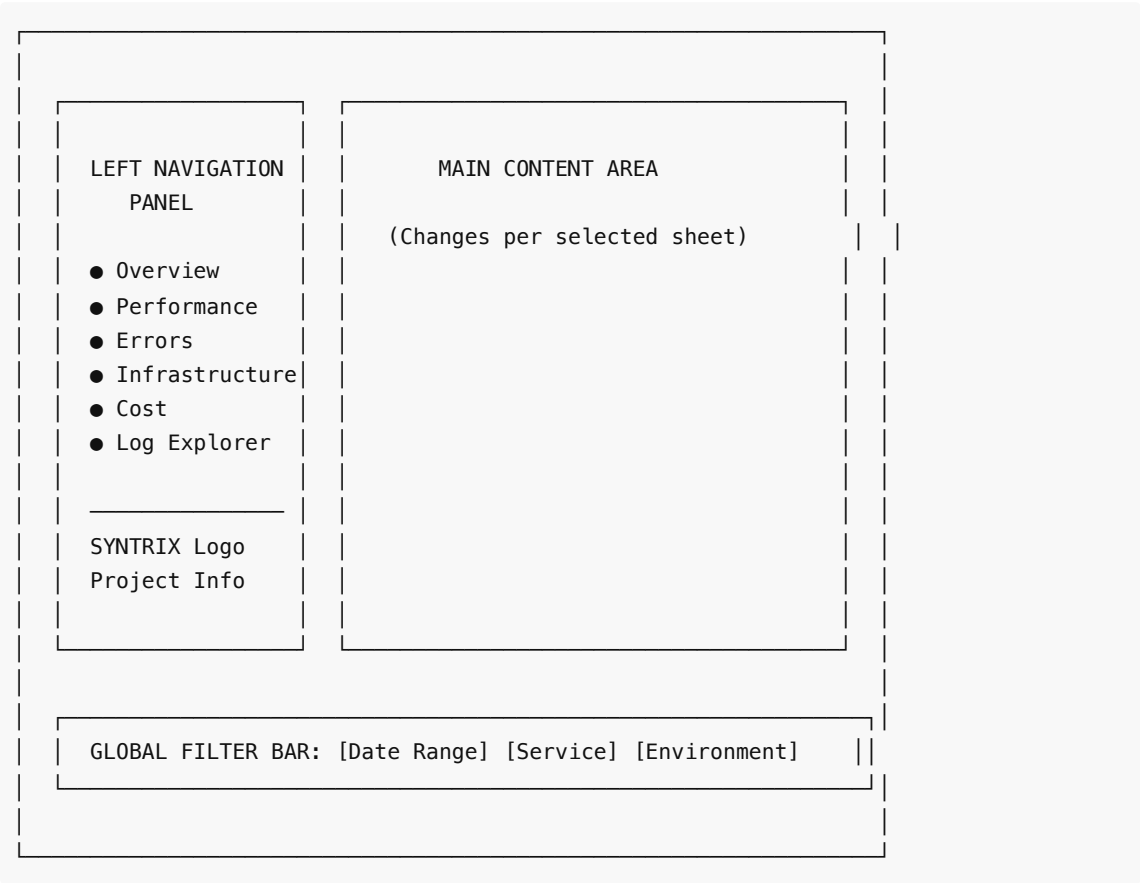
Purpose: Actionable cost-saving recommendations based on 24-hour traffic patterns.

Column	Type	Description
compute_environment	STRING	GCP platform type
service_name	STRING	Service identifier

total_requests_last_24h	INT64	Request count (24h)
cost_recommendation	STRING	Actionable recommendation

5. Looker Studio Dashboard Architecture — Multi-Layer Design

5.1 Dashboard Sheet Map



5.2 Design Principles

Principle	Implementation
Multi-Layer Structure	6 sheets, each targeting a specific stakeholder persona
Left-Side Navigation	Persistent sidebar with icon + label links to each sheet
Responsive Layout	Grid-based layout; works on 1920×1080 and 1366×768
Consistent Branding	Same header, footer, color palette, fonts across all sheets
Filter Persistence	Global date range filter applies across all sheets
Drill-Down Capability	Click a service/endpoint to filter all charts on that sheet

6. Step-by-Step Implementation Guide

PHASE 1 — Setting Up the Looker Studio Report

Step 1: Open Looker Studio

1. Go to <https://lookerstudio.google.com>
2. Sign in with your GCP-linked Google account
3. Click "+ Create" → "Report"

Step 2: Connect the BigQuery Data Sources

You need to add **all 6 BigQuery views** as separate data sources.

1. In the new blank report, click **"Add data"** (bottom toolbar or Resource → Manage added data sources)
2. Select **"BigQuery"** as the connector
3. Navigate: **My Projects** → `project-4e3f1563-833a-4721-bf7` → `syntrix_logs`
4. Select the first view: `looker_cleaned_logs` → Click **"Add"**
5. Repeat for each remaining view:
 - `looker_latency_analysis`
 - `looker_error_analysis`
 - `looker_usage_monitoring`
 - `looker_performance_kpis`
 - `looker_cost_optimization_recommendations`

Beginner Tip: You should now see 6 data sources listed in **Resource → Manage added data sources**. Name each source clearly (e.g., rename `"looker_cleaned_logs"` to `"DS1 - Cleaned Logs"`).

Step 3: Set Up Report Pages (Sheets)

1. At the bottom of the Looker Studio canvas, you will see page tabs
2. Right-click the default page → **"Rename"** → `Executive Overview`
3. Click the "+" icon next to the page tab to add new pages
4. Create 5 more pages and name them:
 - `System Performance`
 - `Error Analysis`
 - `Infrastructure Usage`
 - `Cost Optimization`
 - `Developer Log Explorer`

Step 4: Set Report-Level Theme

1. Go to **Theme and Layout** (top menu bar → "Theme and layout")
2. Click **"Customize"**
3. Set:
 - **Canvas Size:** Custom → `1920 x 1080`
 - **Background Color:** `#0F1724` (dark navy — professional dark theme)
 - **Font Family:** `Roboto` or `Inter`
 - **Primary Text Color:** `#E8ECF1` (light gray-white)
 - **Accent Color 1:** `#4285F4` (Google Blue)
 - **Accent Color 2:** `#34A853` (Google Green)
 - **Accent Color 3:** `#EA4335` (Google Red)

- **Grid spacing:** 10px

PHASE 2 — Building the Left-Side Navigation Panel

This navigation panel must be **copied identically** onto every sheet so it appears persistent.

Step 5: Create the Navigation Sidebar on Sheet 1

1. Draw a Rectangle (Insert → Shape → Rectangle)

- Position: X: 0, Y: 0
- Size: Width: 220, Height: 1080
- Background Color: #151E2D (slightly lighter than canvas)
- Border: None
- Corner Radius: 0

2. Add the Logo / Title at the top of the sidebar

- Insert → Text Box
- Text: < SYNTRIX
- Position: X: 20, Y: 20
- Font: **Roboto Bold**, Size: 22, Color: #4285F4
- Below that, add a subtitle text:
 - Text: CloudMart Monitoring
 - Font: Roboto Regular, Size: 11, Color: #8899AA

3. Add a horizontal divider line

- Insert → Shape → Line
- Position: X: 20, Y: 75 to X: 200, Y: 75
- Color: #2A3545, Thickness: 1px

4. Add Navigation Buttons (one per sheet): Create 6 text boxes, stacked vertically. Each acts as a clickable link to its sheet.

Label	Y Position	Icon	Link To Page
 Executive Overview	Y: 100		Page 1
 Performance	Y: 145		Page 2
 Error Analysis	Y: 190		Page 3
 Infrastructure	Y: 235		Page 4
 Cost Optimization	Y: 280		Page 5
 Log Explorer	Y: 325		Page 6

How to make them clickable:

- Select each text box
- In the Properties panel on the right, scroll down to **"Report Navigation"** or add a **"Link"**
- Set link type to **"Report Page"** → select the corresponding page
- Style: Font size 13, Color #C0CCDA, Padding 8px, Left-align

- **For the ACTIVE page:** Change text color to #4285F4 and add a small 3px wide × 30px tall rectangle in #4285F4 at the left edge of the active button to act as an "active indicator"

5. Add Footer Info at the bottom of the sidebar

- Text: Team SYNTRIX / CTS Hackathon 2026
- Position: near Y: 1020
- Font: Size 10, Color #556677

6. Select All sidebar elements (Ctrl/Cmd + Click each element) → Right-click → "Group" → Copy this Group → Paste onto every other sheet in the exact same position

PHASE 3 — Global Filter Bar

Step 6: Add Filters (on each sheet, below the header)

1. Date Range Control

- Insert → "Date range control"
- Position: X: 240, Y: 20, Width: 250
- Default: "Last 7 Days"
- Style: Background #1A2535, text #E8ECF1, border #2A3545

2. Service Name Filter (Drop-down)

- Insert → "Drop-down list" control
- Data Source: DS1 – Cleaned Logs
- Dimension: service_name
- Position: X: 510, Y: 20, Width: 200
- Label: "Service"

3. Compute Environment Filter (Drop-down)

- Insert → "Drop-down list" control
- Dimension: compute_environment
- Position: X: 730, Y: 20, Width: 200
- Label: "Environment"

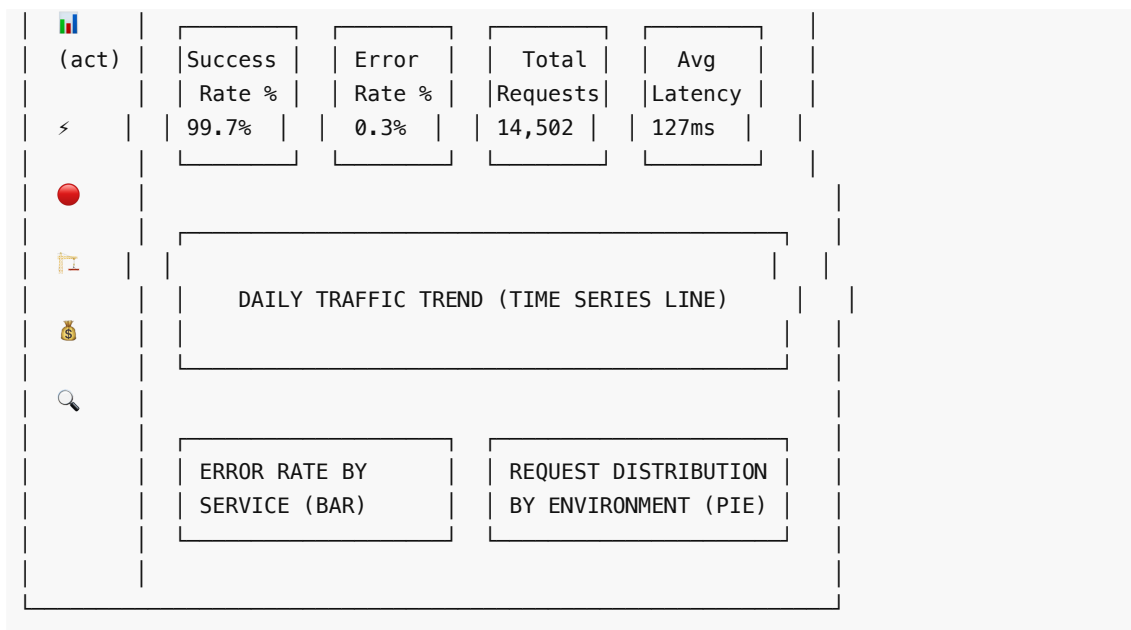
Beginner Tip: These filters automatically cross-filter all charts on the page that use the same data source. If charts use different data sources, you must set up **"Data source filter binding"** under each chart's properties.

7. Sheet 1 — Executive Overview

Data Sources Used: `looker_performance_kpis`, `looker_error_analysis`, `looker_usage_monitoring`

Layout:

[NAV]	HEADER: Executive Overview	[Filters]



Step-by-Step Build:

7.1 KPI Scorecards (Row of 4 cards)

Card 1: Global Success Rate

1. Insert → **Scorecard**
2. Data Source: `looker_performance_kpis`
3. Metric: `success_rate_percentage` → Aggregation: **AVG**
4. Position: X: 250, Y: 90, Size: 280 × 120
5. Style:
 - o Background: `#1A2535`
 - o Value Font: Roboto Bold, Size 40, Color `#34A853` (green)
 - o Label: "Success Rate" — Font size 12, Color `#8899AA`
 - o Border Radius: 12px
 - o Add a shadow for depth
6. Compact Number: OFF
7. Number Format: Percent, 1 decimal place

Card 2: Global Error Rate

1. Insert → Scorecard
2. Data Source: `looker_error_analysis`
3. Metric: `error_rate_percentage` → Aggregation: **AVG**
4. Value Color: `#EA4335` (red)
5. Label: "Error Rate"

Card 3: Total Requests (24h)

1. Insert → Scorecard
2. Data Source: `looker_usage_monitoring`
3. Metric: `daily_requests` → Aggregation: **SUM**
4. Value Color: `#4285F4` (blue)
5. Label: "Total Requests"

Card 4: Average Latency

1. Insert → Scorecard
2. Data Source: `looker_latency_analysis`
3. Metric: `avg_latency_ms` → Aggregation: **AVG**
4. Value Color: `#FBBC04` (yellow/amber)
5. Suffix: "ms"
6. Label: "Avg Latency"

7.2 Daily Traffic Trend Line Chart

1. Insert → **Time Series Chart**
2. Data Source: `looker_usage_monitoring`
3. Date Dimension: `reporting_date`
4. Metric: `daily_requests` → Aggregation: **SUM**
5. Position: Below scorecards, spanning full width (X: 250, Y: 230, Width: 1640, Height: 320)
6. Style:
 - Line Color: `#4285F4`
 - Background: `#1A2535`
 - Grid Lines Color: `#2A3545`
 - Data Point size: 4
 - Enable Trend Line: Yes (linear)
 - X-Axis Label Color: `#8899AA`

7.3 Error Rate by Service (Bar Chart)

1. Insert → **Bar Chart**
2. Data Source: `looker_error_analysis`
3. Dimension: `service_name`
4. Metric: `error_rate_percentage` → Aggregation: **AVG**
5. Position: X: 250, Y: 570, Width: 800, Height: 350
6. Sort: Descending by metric
7. Style: Horizontal bars, Color series `#EA4335` , Background `#1A2535`

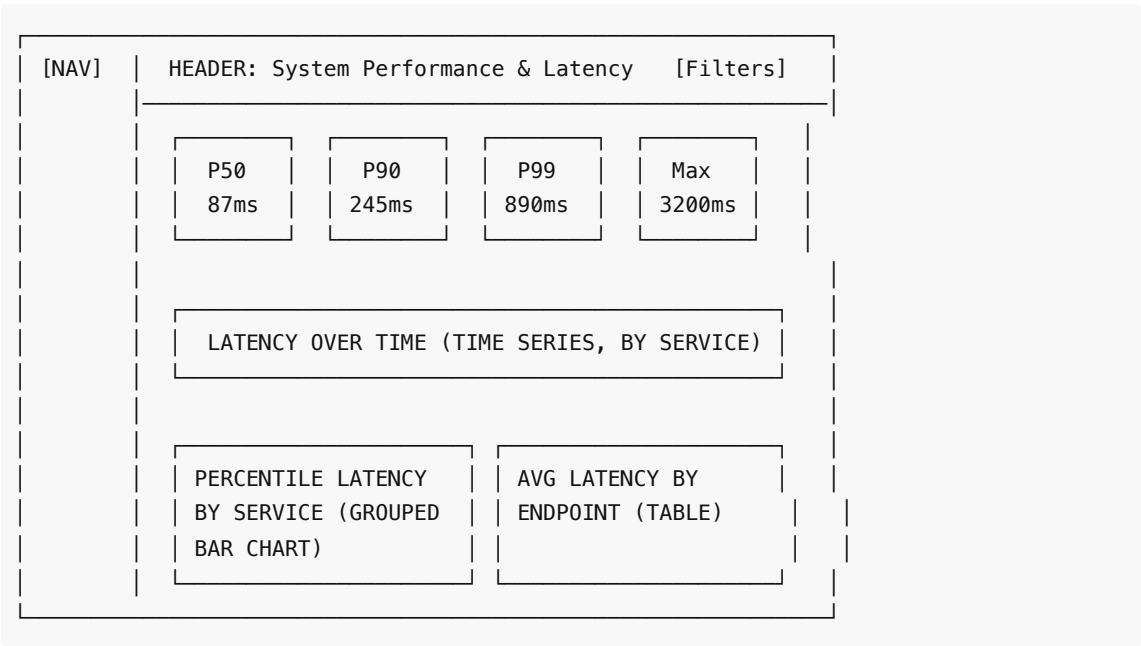
7.4 Request Distribution by Environment (Donut Chart)

1. Insert → **Pie Chart** → Style as **Donut**
2. Data Source: `looker_usage_monitoring`
3. Dimension: `compute_environment`
4. Metric: `daily_requests` → SUM
5. Position: X: 1080, Y: 570, Width: 810, Height: 350
6. Style:
 - Donut Hole Size: 60%
 - Colors: Cloud Run = `#4285F4` , GKE = `#34A853` , GCE = `#FBBC04`
 - Background: `#1A2535`
 - Show Legend: Right side

8. Sheet 2 — System Performance & Latency

Data Sources Used: `looker_latency_analysis` , `looker_performance_kpis`

Layout:



Step-by-Step Build:

8.1 Latency KPI Scorecards

Create 4 Scorecards using data source `looker_performance_kpis` :

Card	Metric	Aggregation	Color
P50	p50_latency	AVG	#34A853
P90	p90_latency	AVG	#FBBC04
P99	p99_latency	AVG	#EA4335
Max	max_latency_ms	MAX (from latency_analysis)	#EA4335

8.2 Latency Over Time (Multi-line Time Series)

- 1. Insert → **Time Series Chart**
- 2. Data Source: `looker_latency_analysis`
- 3. Date Dimension: `time_window`
- 4. Breakdown Dimension: `service_name`
- 5. Metric: `avg_latency_ms` → AVG
- 6. This creates one colored line per service
- 7. Style: Background `#1A2535` , Legend at bottom

8.3 Percentile Latency by Service (Grouped Bar Chart)

- 1. Insert → **Bar Chart** (Grouped/Clustered)
- 2. Data Source: `looker_performance_kpis`
- 3. Dimension: `service_name`
- 4. Metrics: `p50_latency` , `p90_latency` , `p99_latency`
- 5. Bar Colors: P50 = `#34A853` , P90 = `#FBBC04` , P99 = `#EA4335`

6. This gives 3 bars per service, showing the percentile breakdown

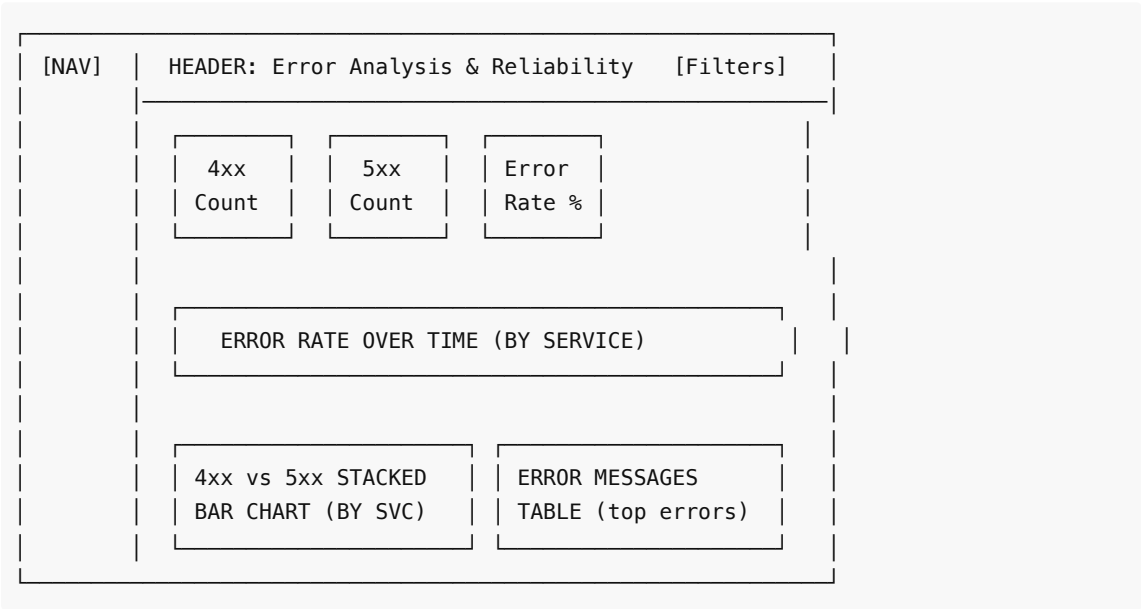
8.4 Latency by Endpoint (Data Table)

- 1. Insert → **Table with Heatmap**
- 2. Data Source: `looker_latency_analysis`
- 3. Dimensions: `service_name` , `endpoint`
- 4. Metrics: `total_requests` , `avg_latency_ms` , `max_latency_ms`
- 5. Sort: `avg_latency_ms` DESC
- 6. Style: Heatmap on `avg_latency_ms` column (green → red gradient)
- 7. Row limit: 25

9. Sheet 3 — Error Analysis & Reliability

Data Sources Used: `looker_error_analysis` , `looker_cleaned_logs`

Layout:



Step-by-Step Build:

9.1 Error KPI Scorecards

Card	Data Source	Metric	Aggregation	Color
4xx Count	looker_error_analysis	client_errors_4xx	SUM	#FBBC04
5xx Count	looker_error_analysis	server_errors_5xx	SUM	#EA4335
Error Rate %	looker_error_analysis	error_rate_percentage	AVG	#EA4335

9.2 Error Rate Over Time (Time Series)

- 1. Insert → **Time Series Chart**
- 2. Data Source: `looker_error_analysis`
- 3. Date Dimension: `time_window`

- 4. Breakdown: `service_name`
- 5. Metric: `error_rate_percentage` → AVG
- 6. Add a **Reference Line** at `5%` (threshold, dashed red) via chart properties

9.3 4xx vs 5xx Stacked Bar (Per Service)

- 1. Insert → **Stacked Bar Chart**
- 2. Data Source: `looker_error_analysis`
- 3. Dimension: `service_name`
- 4. Metrics: `client_errors_4xx` (SUM), `server_errors_5xx` (SUM)
- 5. Colors: 4xx = `#FBBC04` , 5xx = `#EA4335`

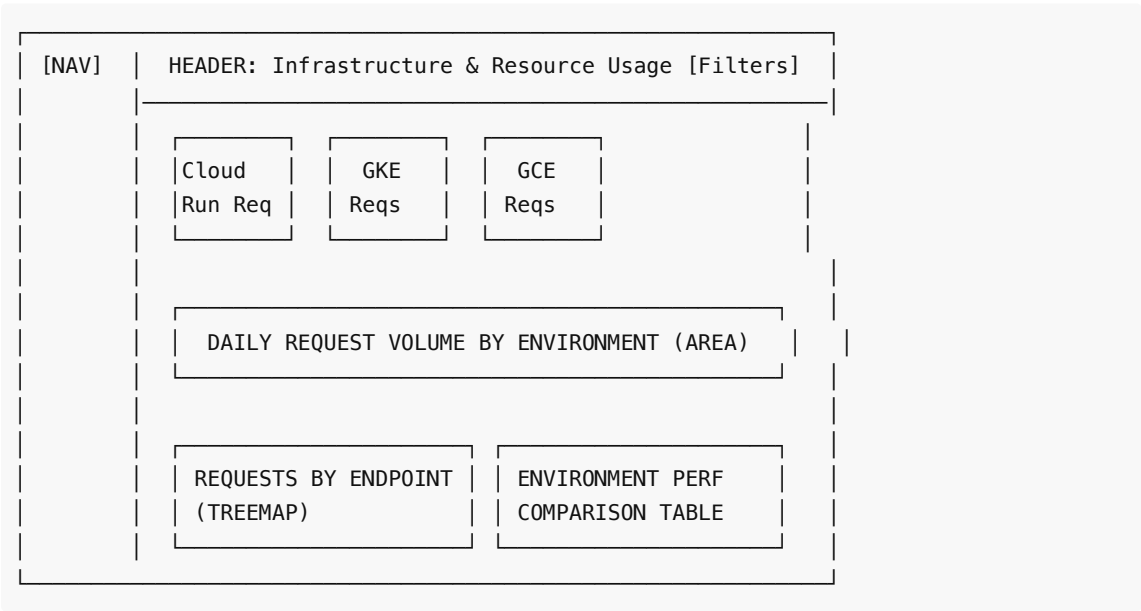
9.4 Top Error Messages Table

- 1. Insert → **Table**
- 2. Data Source: `looker_cleaned_logs`
- 3. Filter: `severity = ERROR` (add chart-level filter)
- 4. Dimensions: `service_name` , `error_message`
- 5. Metric: Record Count
- 6. Sort: Record Count DESC
- 7. Row limit: 15

10. Sheet 4 — Infrastructure & Resource Usage

Data Sources Used: `looker_usage_monitoring` , `looker_latency_analysis`

Layout:



Step-by-Step Build:

10.1 Environment KPI Scorecards

Create 3 scorecards from `looker_usage_monitoring` . Each card uses a **chart-level filter** on `compute_environment` :

Card	Filter	Metric	Color
Cloud Run	compute_environment = cloud_run_revision	daily_requests SUM	#4285F4
GKE	compute_environment = k8s_container	daily_requests SUM	#34A853
GCE	compute_environment = gce_instance	daily_requests SUM	#FBBC04

10.2 Daily Request Volume by Environment (Stacked Area Chart)

- 1. Insert → **Area Chart**
- 2. Data Source: `looker_usage_monitoring`
- 3. Date Dimension: `reporting_date`
- 4. Breakdown: `compute_environment`
- 5. Metric: `daily_requests` → SUM
- 6. Style: Stacked, use environment colors, opacity 70%

10.3 Requests by Endpoint (Treemap)

- 1. Insert → **Treemap Chart**
- 2. Data Source: `looker_usage_monitoring`
- 3. Dimension: `endpoint`
- 4. Metric: `daily_requests` → SUM
- 5. Color scale: Blue gradient

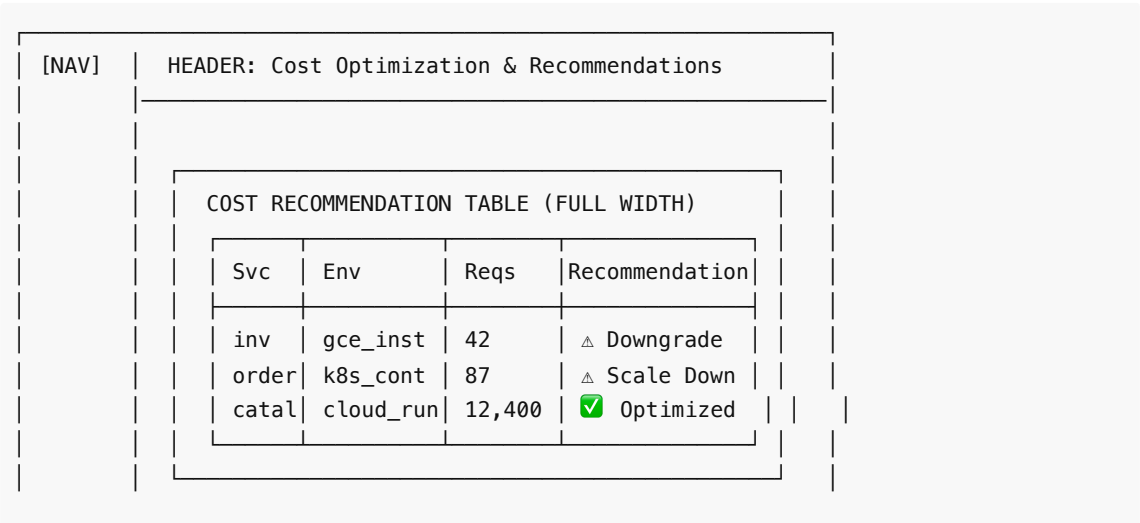
10.4 Environment Performance Comparison Table

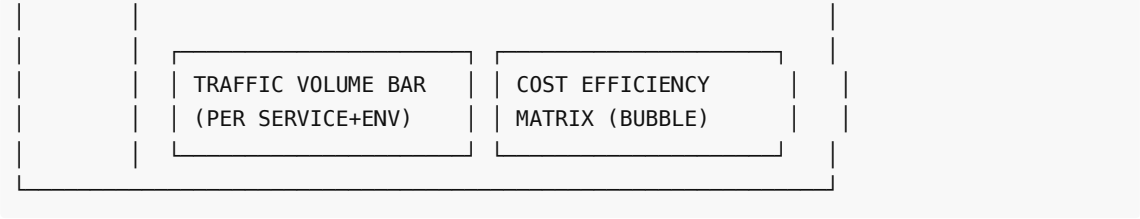
- 1. Insert → **Pivot Table** or **Table with Heatmap**
- 2. Data Source: `looker_latency_analysis`
- 3. Row Dimension: `compute_environment`
- 4. Metrics: `total_requests` SUM, `avg_latency_ms` AVG, `max_latency_ms` MAX
- 5. Heatmap on `avg_latency_ms`

11. Sheet 5 — Cost Optimization & Recommendations

Data Sources Used: `looker_cost_optimization_recommendations` , `looker_usage_monitoring`

Layout:





Step-by-Step Build:

11.1 Cost Recommendation Table (Primary Component)

- 1. Insert → **Table**
- 2. Data Source: `looker_cost_optimization_recommendations`
- 3. Dimensions: `service_name` , `compute_environment` , `cost_recommendation`
- 4. Metric: `total_requests_last_24h` → SUM
- 5. Style:
 - Full width: X: 250, Y: 90, Width: 1640, Height: 360
 - Background: `#1A2535`
 - **Conditional Formatting on `cost_recommendation` :**
 - Contains "Scale down" or "Downgrade" → Background `#FFF3CD` (amber), Text `#856404`
 - Contains "migrate to Cloud Run" → Background `#D4EDDA` (light green), Text `#155724`
 - Contains "Optimized" → Background `#1A2535` , Text `#34A853`
- 6. Sort: `total_requests_last_24h` ASC (show lowest traffic / most actionable first)

11.2 Traffic Volume per Service (Bar Chart)

- 1. Insert → **Bar Chart**
- 2. Data Source: `looker_cost_optimization_recommendations`
- 3. Dimension: `service_name`
- 4. Breakdown: `compute_environment`
- 5. Metric: `total_requests_last_24h` → SUM
- 6. Stacked bars, color-coded by environment

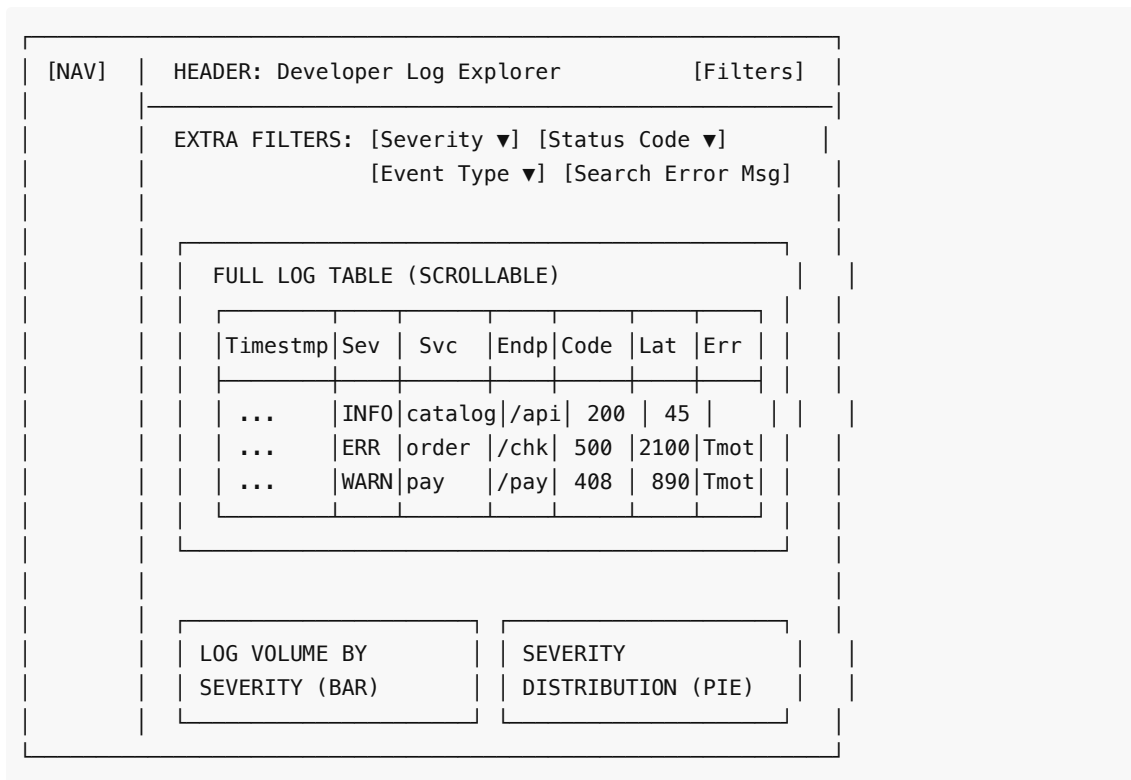
11.3 Cost Efficiency Bubble Chart

- 1. Insert → **Scatter / Bubble Chart**
- 2. Data Source: `looker_latency_analysis`
- 3. Dimension: `service_name`
- 4. X-Axis: `total_requests` (SUM) — represents traffic volume
- 5. Y-Axis: `avg_latency_ms` (AVG) — represents performance
- 6. Bubble Size: `total_requests` (SUM)
- 7. This reveals services with high latency + low traffic (candidates for cost reduction)

12. Sheet 6 — Developer Deep Dive & Log Explorer

Data Sources Used: `looker_cleaned_logs`

Layout:



Step-by-Step Build:

12.1 Additional Filter Controls

Add these filter controls at the top of this sheet only:

Filter	Dimension	Control Type
Severity	severity	Drop-down
Status Code	status_code	Drop-down or Slider
Event Type	event_type	Drop-down
Error Message	error_message	Text Input (search)

12.2 Full Log Table

- 1. Insert → **Table**
- 2. Data Source: `looker_cleaned_logs`
- 3. Dimensions:
 - o `timestamp` (format: `YYYY-MM-DD HH:mm:ss`)
 - o `severity`
 - o `service_name`
 - o `compute_environment`
 - o `endpoint`
 - o `method`
 - o `status_code`
 - o `latency_ms`
 - o `event_type`

- `error_message`
- 4. Sort: `timestamp DESC` (newest first)
- 5. Rows per page: 50
- 6. Enable pagination
- 7. Style:
 - Full width: X: 250, Y: 170, Width: 1640, Height: 550
 - Background: `#1A2535`
 - Header row: `#151E2D` , Bold
 - **Conditional Formatting:**
 - `severity = "ERROR"` → Row background: `rgba(234,67,53,0.15)` (faint red)
 - `severity = "WARNING"` → Row background: `rgba(251,188,4,0.15)` (faint amber)
 - `status_code >= 500` → Cell `#EA4335` bold

12.3 Log Volume by Severity (Bar Chart)

- 1. Insert → **Bar Chart**
- 2. Dimension: `severity`
- 3. Metric: Record Count
- 4. Colors: INFO = `#4285F4` , WARNING = `#FBBC04` , ERROR = `#EA4335`

12.4 Severity Distribution (Donut Chart)

- 1. Insert → **Pie Chart** (Donut style)
- 2. Dimension: `severity`
- 3. Metric: Record Count
- 4. Same color scheme as above

13. Left-Side Navigation Panel Design

13.1 Visual Specification

SYNTRIX

CloudMart Monitor

Overview

Performance

Errors

Infrastructure

Cost

Log Explorer

← Logo, color #4285F4

← Subtitle, color #8899AA

← Divider, color #2A3545

← Active: blue indicator bar + #4285F4 text

← Inactive: #C0CCDA text

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CTS Hackathon '26

← Footer, color #556677

Width: 220px

13.2 Interactivity

- Each label is a **Text Box** with a **Report Page link** configured
- On hover: Text color changes (configure via Looker Studio text style)
- The "active indicator" (small blue bar) is manually placed on each sheet to highlight the current active page

13.3 Replication Across Sheets

1. Build the sidebar on Sheet 1
2. Group all elements (Ctrl/Cmd+A within the sidebar area → Right-click → Group)
3. Copy → Navigate to Sheet 2 → Paste in Place (Ctrl/Cmd+Shift+V)
4. On each sheet, update the active indicator position to match the current page
5. Repeat for all 6 sheets

14. Responsive Design & Theme Guidelines

14.1 Color Palette

Token	Hex	Usage
Canvas Background	#0F1724	Report background
Card Background	#1A2535	All chart/card backgrounds
Sidebar Background	#151E2D	Navigation panel
Divider / Border	#2A3545	Subtle separators
Primary Text	#E8ECF1	Headings, values
Secondary Text	#8899AA	Labels, subtitles
Muted Text	#556677	Footer, disabled state
Accent Blue	#4285F4	Primary actions, active states
Success Green	#34A853	Positive KPIs, success states
Warning Amber	#FBB03B	Warnings, moderate values
Error Red	#EA4335	Errors, critical alerts

14.2 Typography

Element	Font	Size	Weight	Color
Page Title	Roboto	24px	Bold	#E8ECF1

Section Title	Roboto	16px	Medium	#E8ECF1
Card Value	Roboto	36-40px	Bold	Accent
Card Label	Roboto	12px	Regular	#8899AA
Table Header	Roboto	12px	Bold	#C0CCDA
Table Body	Roboto	11px	Regular	#E8ECF1
Nav Label	Roboto	13px	Medium	#C0CCDA

14.3 Layout Grid

Property	Value
Canvas Width	1920px
Canvas Height	1080px
Sidebar Width	220px
Content Area Start	X: 250px
Content Area Width	1640px
Grid Spacing	10px
Card Border Radius	12px
Chart Padding	16px
Gap between cards	20px

14.4 Responsive Considerations

- Looker Studio auto-scales on smaller screens via its responsive setting
- Go to **File** → **Report Settings** → **Layout**: set to "Fit to width"
- Test at 1920×1080 and 1366×768 by resizing the browser window
- Avoid placing critical elements beyond X: 1800 to prevent overflow

15. Alert Integration & Threshold Configuration

15.1 GCP Cloud Monitoring Alerts (Already Configured)

These alerts are already deployed via `create_alerts.sh` :

Alert Name	Condition	Recipient
Cloud Run 5xx Errors	5xx count > 10 in 60s	Babu (Cloud Run Owner)
GKE High CPU	CPU utilization > 80% for 3min	Akash (GKE Owner)
GCE High CPU	CPU utilization > 85% for 3min	Roshni & Swathi (GCE)
BigQuery High Execution Time	Query time > 10s in 60s	Ecclesiastes & Emayan (BQ)

Log Streamer Pipeline Errors	ERROR logs > 5 in 60s	Varunshiyam (Pipeline)
------------------------------	-----------------------	------------------------

15.2 Looker Studio Scheduled Email Delivery

1. Go to **Share** → **Schedule email delivery**
2. Set frequency: **Daily at 9:00 AM IST**
3. Recipients: Team distribution list
4. Subject: [SYNTRIX] Daily CloudMart Dashboard Report
5. Select which pages to include (recommend: Executive Overview + Cost Optimization)

15.3 Embedding Alerts Context in Dashboard

- On Sheet 1 (Executive Overview), add a **Text Box** at the bottom titled "**Active Alert Policies**"
- List the alert thresholds from section 15.1 as reference info for stakeholders
- This provides context when a scorecard shows a degraded metric

16. Appendix — LookML Reference (If Using Looker Enterprise)

Note: This section is relevant only if you are using **Looker Enterprise** (not Looker Studio). Looker Studio uses direct BigQuery connections. Looker Enterprise requires LookML model definitions.

16.1 Model File (`syntrix.model.lkml`)

```
connection: "syntrix_bigquery"

include: "/views/*.view.lkml"

explore: cleaned_logs {
  label: "CloudMart Log Explorer"
  description: "Base cleaned log data for all services"
}

explore: latency_analysis {
  label: "Latency Analysis"
  description: "Hourly latency aggregations by service and endpoint"
}

explore: error_analysis {
  label: "Error Analysis"
  description: "Hourly error breakdown with 4xx/5xx split"
}

explore: performance_kpis {
  label: "Performance KPIs"
  description: "Percentile latency and success rates"
}

explore: cost_optimization {
  label: "Cost Optimization"
```

```
description: "Cost-saving recommendations based on traffic"
}
```

16.2 Example View File (cleaned_logs.view.lkml)

```
view: cleaned_logs {
  sql_table_name: `project-4e3f1563-833a-4721-bf7.syntrix_logs.looker_cleaned_logs`
;;

  dimension_group: timestamp {
    type: time
    timeframes: [raw, time, hour, date, week, month]
    sql: ${TABLE}.timestamp ;;
  }

  dimension: severity {
    type: string
    sql: ${TABLE}.severity ;;
  }

  dimension: service_name {
    type: string
    sql: ${TABLE}.service_name ;;
  }

  dimension: compute_environment {
    type: string
    sql: ${TABLE}.compute_environment ;;
  }

  dimension: endpoint {
    type: string
    sql: ${TABLE}.endpoint ;;
  }

  dimension: method {
    type: string
    sql: ${TABLE}.method ;;
  }

  dimension: status_code {
    type: number
    sql: ${TABLE}.status_code ;;
  }

  dimension: latency_ms {
    type: number
    sql: ${TABLE}.latency_ms ;;
  }

  dimension: event_type {
```

```

    type: string
    sql: ${TABLE}.event_type ;;
}

dimension: error_message {
  type: string
  sql: ${TABLE}.error_message ;;
}

measure: total_requests {
  type: count
  drill_fields: [timestamp_time, service_name, endpoint, status_code, latency_ms]
}

measure: avg_latency {
  type: average
  sql: ${latency_ms} ;;
  value_format: "0.0 \"ms\""
}

measure: error_rate {
  type: number
  sql: 1.0 * ${error_count} / NULLIF(${total_requests}, 0) * 100 ;;
  value_format: "0.00 \"%\""
}

measure: error_count {
  type: count
  filters: [status_code: ">=400"]
}

measure: success_rate {
  type: number
  sql: 100.0 - ${error_rate} ;;
  value_format: "0.00 \"%\""
}
}

```

Quick Reference Checklist

Use this checklist to track your progress:

- ☐ **Phase 1:** Looker Studio report created
- ☐ **Phase 1:** All 6 BigQuery data sources connected
- ☐ **Phase 1:** 6 pages/sheets created and named
- ☐ **Phase 1:** Dark theme applied
- ☐ **Phase 2:** Left-side navigation sidebar built on Sheet 1
- ☐ **Phase 2:** Navigation links configured to each page
- ☐ **Phase 2:** Sidebar copied to all 6 sheets

- ☐ **Phase 3:** Global filter bar added (Date, Service, Environment)
- ☐ **Sheet 1:** Executive Overview — 4 KPI cards + 2 charts
- ☐ **Sheet 2:** System Performance — Latency scorecards + time series + bar + table
- ☐ **Sheet 3:** Error Analysis — Error scorecards + time series + stacked bar + error table
- ☐ **Sheet 4:** Infrastructure Usage — Environment cards + area chart + treemap + table
- ☐ **Sheet 5:** Cost Optimization — Recommendation table + bar chart + bubble chart
- ☐ **Sheet 6:** Log Explorer — Extra filters + full log table + severity charts
- ☐ **Final:** Responsive layout tested at 1920×1080 and 1366×768
- ☐ **Final:** Scheduled email delivery configured
- ☐ **Final:** Report shared with team

End of Document

Prepared by SYNTRIX Analytics Team — CloudMart Enterprise Observability Platform